Amaan Mansuri

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Education

New York University – Center for Data Science | New York, USA
Master of Science in Data Science | *Expected May 2026*

Nirma University – Institute of Technology | Ahmedabad, India
Bachelor of Technology in Computer Science and Engineering | *Sept 2020 - Jun 2024*

Experience

Data Science Intern (NLP) | New York, USA
Lepercq de Neuflize Asset Management LLC | *Feb 2026 – Present*

- Built an end-to-end **NLP data pipeline** for stylometric analysis of hedge fund investor letters, covering ingestion, cleaning, sectioning, **text classification**, feature extraction (lexical/syntactic/semantic), and reproducible reporting.
- Engineered author and period-level signals (e.g., **style consistency**, readability, **sentiment analysis**, stance, topic drift) to support qualitative investment research and **flag shifts in communication patterns** across managers.
- Automated a workflow comparing letters across **15+ funds** and multi-year horizons, producing advisor-ready summaries that reduced manual text triage time by **60%**.

Research Assistant (NLP) | New York, USA
Hartley Lab, Department of Psychology, New York University | *Mar 2025 – Present*

- Developed a web scraping pipeline to collect and process over **15,000** metadata entries from books and media sources, cutting manual stimuli search time by **60%**.
- Proposed and implemented **knowledge graph-based** models using **NetworkX** to simulate semantic relationships across age groups, boosting prediction accuracy in reasoning tasks by **25%** over baseline vector-similarity methods.
- Designed story-based experimental tasks and built data processing workflows for behavioral datasets, supporting research bridging **cognitive neuroscience** and **computational modeling**.

AI/ML Research Engineer (CV) | New York, USA
United Nations Development Programme (UNDP) | *Sep 2025 – Dec 2025*

- Designed and deployed an **underwater fish detection pipeline** for marine biodiversity monitoring in **production**: dataset curation, QA workflows, **feature engineering**, and robust preprocessing for severe underwater challenges (extreme blue tint, low contrast, turbidity).
- Achieved **70.1% mAP50** via **5-model Weighted Boxes Fusion** ensemble, exceeding target accuracy under a 70MB edge-deployment constraint; benchmarked YOLOv11 variants across **60+** hyperparameter experiments.
- Integrated **multi-object tracking** into the detection pipeline to enable real-time species monitoring via underwater video, supporting both ecological research and **VR-based tourism** without habitat disruption.

Projects

GEPA for LLM Reliability (Research Project) | *Github* *Blog*

- Developed a reflective **prompt engineering** system using GEPA methodology to improve **generative AI** reliability on fact-checking (HoVer) and code generation (HumanEval); achieved **100% Pass@1 on HumanEval** for GPT-3.5 Turbo (from 79.69%) and **+32.81pp** for Qwen 3-8B at Pass@5.
- Extended GEPA framework with novel few-shot-aware variant and conducted systematic multi-model analysis across **transformer**-based LLMs (GPT-4.1 mini, GPT-3.5 Turbo, Qwen 3-8B), demonstrating cross-domain generalizability with only **5 iterations** and no **fine-tuning** required.

flight-wx: Flight Delay & Weather Monitoring System (Big Data + API Integration) | *Github*

- Developed a real-time flight delay tracking system integrating FAA and NOAA APIs, supporting flexible city-to-airport queries and resolving metadata for **400+ U.S. airports** with high accuracy and reliability.
- Engineered modular **ETL** tools with **REST API** integration, batch-mode ingestion, intelligent caching, and multi-airport resolution logic, reducing redundant API hits by **70%** and improving end-to-end ingestion speed by **3x**.

Multi-Horizon Stock Price Prediction (LSTM + FinBERT) | *Github* *Report*

- Built a deep learning pipeline with **feature engineering** across **LSTM networks** to forecast 1-day and 3-day prices for **26 major stocks**, integrating **FinBERT sentiment scores** to boost accuracy by up to **18.57%** and directional accuracy by **12%**.
- Evaluated four feature sets (price, technicals, sentiment, combined) across standard regression metrics; achieved best-case **R^2 = 0.969** (AAPL, 1-day) with `ReduceLROnPlateau` and `EarlyStopping` scheduling.

Skills

Languages: Python, SQL, Java, C++, Bash.
ML/NLP: PyTorch, TensorFlow, JAX, scikit-learn, Hugging Face, Transformers, OpenCV, NLTK, spaCy, LangChain, RAG.
Data & Infrastructure: NumPy, Pandas, Spark, Dask, Docker, Airflow, MLflow, AWS, GCP, FastAPI, REST APIs, Git.

Publications & Awards

Multilingual Handwritten Digit Recognition Using Multiplexer-Based DL Models: IEEE – 2024 15th ICCCNT
High-Precision Estimation of DES Density and Viscosity using XGBoost: Springer – Journal of Solution Chemistry
Violet Internship & Research Award: NYU; recognizes outstanding contributions to faculty-led research